IB COMPUTER SCIENCE IA — COMPLETE STUDENT DOCUMENTATION GUIDE

A Full Start-to-Finish Blueprint for Documentation, Process, and Deliverables

(Based strictly on IB requirements, moderator feedback, and the official handbook)

SECTION 0 — OVERVIEW OF THE IA

The IA is a **complete software development project** demonstrating real-world problem solving, computational thinking, and programming complexity. You must show:

- A **real client** with a real problem
- **Evidence of consultation** with that client
- A carefully planned **design cycle**
- A **non-trivial, technically robust product**
- A **clear video** showing the product meets every success criterion
- A **thorough evaluation** with client feedback

Your final submission consists of:

Criterion	Deliverable	Format	Word Count
A	Planning (Scenario, Rationale, Success Criteria)	PDF	~500 total
B	Record of Tasks	IB Template	none
B	Design Overview	PDF	none
C	Development	PDF	500–1000
D	Video	MP4	4–7 min
D	Extensibility	Evidenced in B & C	none
E	Evaluation + Recommendations	PDF	~500
—	Appendix	PDF	none

SECTION 1 — DOCUMENT STRUCTURE (RECOMMENDED)

Below is the best, safest, most IB-aligned structure.
Students should create one master Google Doc or Word file at the beginning containing the following sections, then later **export each section separately as a PDF for submission**, exactly as IB requires.

SECTION 2 — WHEN TO COMPLETE EACH DOCUMENT

This is crucial — IB examiners look for evidence of **process**, not retroactive fill-in. Students should follow this timeline:

IA Stage	Student Action	Documents to Fill In
Week 1: Client Selection & Problem Investigation	Conduct Interview #1, gather data	- Start Criterion A Scenario draft - Add Interview #1 transcript to Appendix
Week 2: Define Rationale & Preliminary Success Criteria	Conduct Interview #2 to validate criteria	- Finalize A: Scenario, Rationale, Success Criteria - Add Interview #2 transcript to Appendix
Week 3–4: System Planning	Begin structured design	- Record of Tasks (start filling, ongoing) - Design Overview (start diagrams, mockups, data structures)
Week 5–9: Coding & Development	Build product in small iterations	- Record of Tasks (keep up to date) - Criterion C (write as you code) - Screenshot evidence
Week 10: Final Testing	Execute full test plan	- Finalize Test Plan (Criterion B) - Prepare test evidence
Week 11: Film Video	Demonstrate all success criteria	- Criterion D Video
Week 12: Evaluation	Evaluate solution using client feedback	- Criterion E Evaluation & Recommendations - Add client evaluation interview in Appendix
Final Week	Produce PDFs & package IA	- Create final PDFs - Prepare Appendix - Prepare product folder - Fill IA cover sheet (index.html)

SECTION 3 — DETAILED DOCUMENTATION GUIDE BY CRITERION

⭐ CRITERION A — PLANNING (6 marks)

Deliverables:

1. **Scenario** *(175–250 words)*
2. **Rationale** *(175–250 words)*
3. **Success Criteria** *(bullet points only — no extended text)*

PART A.1 – The Scenario

This is the single biggest determinant of whether the IA succeeds or fails.
A weak or contrived client → lower marks across *all* criteria.
(Moderator note: "contrived tasks lead to weaker projects"

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What the Scenario MUST include

Use this structure:

(1) Identify the Client

- Real human being (teacher, coach, business owner, student leader, etc.)
- Must be able to meet with you twice.
- Should have real needs & be able to evaluate the product.

(2) Describe Their Current Problem

Examiners want:

- **Not a product description**
- **Not "they need an app"**
- You must describe the **real-world inefficiency**, **pain point**, or **workflow issue**.

(3) Explain Limitations of the Current Approach

Quote the client directly:
"The spreadsheets we use often break when new members join..."
"Currently I track attendance manually, which leads to errors..."

(4) Why Software is Needed

Clearly tie the problem to a need for a computational solution.

(5) Evidence of Consultation

You MUST reference the interviews directly in the scenario or you cannot score a 6.

Example required sentences:

- *"During my first consultation with the client (Appendix A1), they explained that…"*
- *"We agreed that the primary problem is…"*

PART A.2 – The Rationale

(175–250 words)
The rationale answers:
Why is this the right solution?
Why is your chosen tech stack the right one?

Required components

(1) Justify the type of product

Example:

- *"The client requested the ability to scan QR codes; therefore a mobile solution is necessary…"*

(2) Justify programming language and technology stack

Moderator note: "Software choices must be justified or max mark is 5."

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Include:

- programming language
- frameworks/libraries
- storage method
- hosting method (if applicable)

(3) Evidence of client approval

End with:
"The client reviewed and approved this proposed direction (Appendix A2)."

PART A.3 – Success Criteria

These must be:

✔ Specific
✔ Measurable
✔ Testable
✔ Directly demonstrable in the video (Criterion D)

Examples of strong success criteria:

- *“The system must allow the client to create, edit, and delete event entries, with each operation updating the persistent SQLite database.”*

- *“The product must generate downloadable CSV attendance files including date, student ID, and attendance code.”*

Aim for 8–12 criteria
Moderator note: “A range is required for top marks.”

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⭐ CRITERION B — RECORD OF TASKS + DESIGN OVERVIEW (6 marks)

PART B.1 – Record of Tasks

You MUST use the IB template (zero marks awarded if missing).
See template on page 10.

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How to fill it out

Each row should include:

- Task number
- Planned action
- Planned outcome
- Time estimate
- Completion date
- Criterion (the stage: Plan / Design / Develop / Test / Implement)

Tips for high performance

- Update continuously (IB examiners can detect retroactive completion).
- Include **implementation tasks**, such as installing the product for the client.
- Avoid tasks like “write Criterion A”—this is incorrect. The ROT is for building the product.

PART B.2 – Design Overview

This is the **second biggest differentiator between high and low scoring IAs.**

Moderator note: “A complete design must include detailed algorithms and a description of data structures”

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Your design overview must include:

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(1) System Architecture Diagram

Depending on your project type:

- Flow of data
- Input → Processing → Storage → Output
- External services (APIs, databases)

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(2) User Interface Mockups

Tools recommended (page 11): Adobe XD, MockFlow, Moqups.

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Include:

- Every major screen
- Annotation of *functionality* not aesthetics

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(3) Data Structures

Examples:

- Class definitions
- Object attributes
- ER diagrams (if using SQL)
- JSON structure
- Arrays / lists / dictionaries
- Relationships

IB expects sophistication here.

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(4) Algorithms (Detailed)

This is where many students miss out on marks.

Provide:

- Pseudocode
- Flowcharts

- UML
- Annotations explaining decisions

What should be included:

✔ Sorting / Searching
✔ Calculations
✔ Validation logic
✔ File/database interactions
✔ Key computational workflows (ex: QR code scanning, form submission process)

IB examiners are looking for **algorithmic thinking**, not just UI design.

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(5) Test Plan

Your test plan must include:

Action	Method	Expected Outcome	Test Data
Normal case			
Abnormal case			
Extreme case			

Test must directly map to **every success criterion**.

⭐ CRITERION C — DEVELOPMENT (12 marks)

This is the **largest mark band** and the place where students lose the most points.

What Criterion C must provide

✔ Explanation of how the product was developed

✔ Justification for why each technique was appropriate

✔ Screenshot evidence

✔ Algorithmic thinking

✔ Explanations of complex techniques

✔ References to any external code

Moderator note: A narrative alone is not enough; you must explain, justify, annotate.

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STRUCTURE FOR CRITERION C

Use this structure:

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(1) Product Structure Overview

Describe:

- Folder structure
- Major modules
- Data flow
- How files/classes interact

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(2) Explanation of Each Major Feature

For each major part:

- What it does
- Why you designed it this way
- Screenshot of code
- Annotation explaining:

✔ input validation
✔ algorithmic steps
✔ data structure use
✔ efficiency considerations
✔ error handling

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(3) Demonstration of Techniques (Complexity + Ingenuity)

High-band techniques include (page 15):

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- Persistent storage (SQL, files)
- APIs
- Dynamic data structures
- OOP
- Event-driven programming
- Asynchronous operations
- MVC separation
- Error handling
- Multi-language systems (e.g., Python backend + JS frontend)

Students should create:

A bulleted list of techniques used, each with:

- screenshot
- short explanation
- why appropriate

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(4) Evidence of Algorithmic Thinking

This should include:

- Searching
- Sorting
- Parsing
- Validation
- Data transformation
- Calculation
- Aggregation
- Conditional logic

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(5) References to external material

Every StackOverflow snippet or library must be cited.

Failure to cite → loss of marks + risk of malpractice.

⭐ CRITERION D — FUNCTIONALITY VIDEO + EXTENSIBILITY (4 marks)

PART D.1 – Functionality Video

Length: 5–7 minutes
Format: MP4 recommended

Structure of the video (recommended)

(1) Title screen:
Project title, candidate code, client name

(2) Show each success criterion one by one
Narrate:

- "Success Criterion 1: The system must…"
- Show it working, including:
  ✔ normal case
  ✔ abnormal case
  ✔ data persistence

Moderator note: "Students often highlight instead of demonstrate functionality."

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(3) Edge cases
IB wants to see **error handling**.

(4) Show data changes
For example:

- A record is added → check the database file
- A calculation is performed → show correct output
- A file is generated → open it

(5) Last 30 seconds: Extensibility
Discuss:

- How files, classes, naming conventions make it maintainable
- How a developer could extend the system

PART D.2 – Extensibility

Demonstrated through:

- Clean folder structure
- Readable code
- Comments
- Naming conventions
- Data structure choices
- Documentation in Criterion C

⭐ CRITERION E — EVALUATION & RECOMMENDATIONS (6 marks)

PART E.1 – Evaluation

This must:

✔ Go through **every** success criterion
✔ Reference **client feedback directly**
✔ Include quotes from client
✔ Include transcript of client evaluation interview in Appendix
✔ Mention tester feedback (if applicable)

Moderator note: “Just stating that feedback exists is not enough; quoting is required.”

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Structure:

For each success criterion:

- State the criterion
- Describe the evidence
- Describe the client’s evaluation
- State whether it was fully/partially/not met

PART E.2 – Recommendations (realistic, justified)

Not allowed:

- “Improve UI” (too trivial)
- “Add more features” (vague)
- “Fix bugs” (unacceptable)

Allowed:

- New structural improvements
- Additional modules expanding functionality
- Performance tuning
- Alternative data storage

- Network capabilities (only if reasonable)

Moderator note: “Recommendations must go beyond unfulfilled success criteria.”

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⭐ APPENDIX STRUCTURE (Required)

Must include:

✔ Interview #1 transcript
✔ Interview #2 transcript
✔ Client evaluation interview
✔ Code listing (full source code)
✔ Additional screenshots
✔ Additional design mockups
✔ Additional test results

Nothing in the appendix is credited unless referenced in the main text.

⭐ FINAL PACKAGING AND SUBMISSION (Critical)

Follow the instructions from pages 25–26 of the handbook:

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Mandatory folder structure:

```
/ProjectRoot
   /Documentation
      Crit_A_Planning.pdf
      Crit_B_Record_of_tasks.pdf
      Crit_B_Design.pdf
      Crit_C_Development.pdf
      Crit_D_Functionality.mp4
      Crit_E_Evaluation.pdf
      Appendix.pdf
   /Product
      (your program files)
index.html
```

index.html Must Include:

- Candidate code
- Solution title
- Links to product (online + offline if applicable)
- Directions for running product
- Word count

Finally:

- Zip folder
- Rename to your candidate code (ex: abc123.zip)

⭐ BEST END PRESENTATION FORMAT

The most professional and IB-safe format:

All documents: PDF exported from Google Docs/Word

All diagrams: PNG embedded inside PDFs

Video: MP4

Product: native code + compiled version